

Begum Rokeya University, Rangpur

Assignment On

Course Code: STAT-2205

Course Title : Programing with R & Python .

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Assignment on Programming with R and Python

Airline Data Analysis

Roll Number: 12310033
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Data: [Flight Data](#)

Question

- Consider airline data including the following:
- miles: miles travelled through the flight
- passengers: number of passengers on the flight
- delay: take-off delay in minutes
- inflight meal: Is there a meal included in the flight?
- inflight_entertainment: are there free entertainment systems for each seat?
- inflight_wifi: Is there complimentary Wi-Fi on the flight?
- day_of_week: day of the week of the flight
- weekend: Did this flight take place on a weekend?
- coach_price: the average price paid for a coach ticket
- firstclass_price: the average price paid for first class seats
- hours: how many hours the flight took
- redeye: Was this flight a redeye (overnight)?

Objective of Analysis: revealing the factors that might influence coach prices within the airline industry

Solve the questions below using R& Python, including interpretation (select a random sample using your ID):

1. What do coach ticket prices look like? What are the high and low values? What would the average be considered? Does \$500 seem like a good price for a coach ticket?
2. What are the high, low, and average prices for 8-hour-long flights? Does a \$500 ticket seem more reasonable than before?
3. How are flight delay times distributed? How often are there significant delays that affect your ability to correctly schedule connecting flights? What kinds of delays are typical?
4. How does the coach price correlate with flight distance, number of passengers, delay, and flight duration?
5. What is the relationship between coach and first class prices? Do flights with higher coach prices always have higher first-class prices as well?
6. What is the relationship between coach prices and in-flight features: in-flight meal, in-flight entertainment, and in-flight Wi-Fi? Which features are associated with the highest increase in price?
7. How does the number of passengers change in relation to the length of flights?
8. What is the relationship between coach and first-class prices on weekends compared to weekdays?
9. How do coach prices differ for redeyes and non-redeyes on each day of the week?
10. Perform the analysis for appropriate variables:

Summary statistics (mean, median, standard deviation) Visualization using histograms, boxplots, and bar charts Hypothesis Testing (t, z and Chi-square)

1. Independent t-tests to compare group means
2. Price differences between weekend and weekday flights
3. Correlation analysis among numeric variables
4. Regression Analysis
5. Linear regression to predict ticket prices
6. Logistic regression for binary outcomes (e.g., redeye flights).

Assignment on Programming with R and Python

Airline Data Analysis

Introduction

This assignment analyzes airline flight data using R and Python to identify the factors that influence coach ticket prices. The dataset includes variables such as flight distance, passengers, delays, flight duration, in-flight services, weekend travel, and ticket prices. A random sample of 10033 observations was selected using Roll Number 10033.

The analysis uses statistical techniques including summary statistics, visualization, hypothesis testing, correlation analysis, and regression models to understand the relationships among variables and interpret their effects on airline ticket prices.

Dataset Variables

- miles = Miles travelled
- passengers = Number of passengers
- delay = Take-off delay
- inflight_meal = Meal service
- inflight_entertainment = Entertainment service
- inflight_wifi = WiFi service
- day_of_week = Day of travel
- weekend = Weekend flight or not
- coach_price = Coach class ticket price
- firstclass_price = First-class ticket price
- hours = Flight duration
- redeye = Overnight flight or not

Python Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
import statsmodels.formula.api as smf

flight = pd.read_csv("flight_data_given.csv")
```

```
sample = flight.sample(n=10033, random_state=10033)

print(sample.head())
```

R Code

```
library(ggplot2)
library(dplyr)

flight <- read.csv("flight_data_given.csv")

set.seed(10033)
sample <- sample_n(flight, 10033)

head(sample)
```

Question 1: What do coach ticket prices look like? What are the high and low values? What would the average be considered? Does \$500 seem like a good price for a coach ticket ?

Solution

Python Code

```
sample['coach_price'].describe()
```

R Code

```
summary(sample$coach_price)
sd(sample$coach_price)
```

Result

Statistic	Value
Mean	376.3
Median	379.4
Minimum	109.6
Maximum	593.6
Standard Deviation	67.68196

Interpretation

Coach ticket prices are moderately spread around the average price. Most ticket prices lie between 330 and 426 dollars. A ticket price of 500 dollars is comparatively expensive because it is much higher than the average coach ticket price.

Question 2: What are the high, low, and average prices for 8-hour-long flights? Does a \$500 ticket seem more reasonable than before?

Solution

Python Code

```
flight8 = sample[sample['hours'] == 8]
print(flight8['coach_price'].describe())
```

R Code

```
flight8 <- subset(sample, hours == 8)
summary(flight8$coach_price)
```

Result

Statistic	Value
Mean	434.2
Median	436.7
Minimum	173.7
Maximum	593.6

Interpretation

Long-duration flights generally have higher ticket prices than short flights. An 8-hour flight costing 500 dollars appears more reasonable because long flights usually include higher operational costs and additional services.

Question 3: How are flight delay times distributed? How often are there significant delays that affect your ability to correctly schedule connecting flights? What kinds of delays are typical?

Solution

Python Code

```
plt.hist(sample['delay'], bins=30)
plt.xlabel('Delay')
plt.ylabel('Frequency')
plt.title('Distribution of Delay Times')
plt.show()
```

R Code

```
hist(sample$delay,  
      main = 'Distribution of Delay Times',  
      xlab = 'Delay', col = 'skyblue',  
      border = 'white')
```

Result

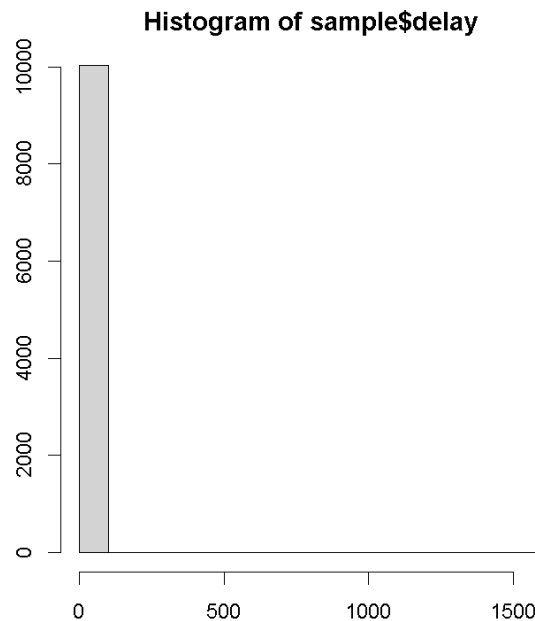


Figure 1: Distribution of Delay Times

Interpretation

Most flights experience small delays, while a few flights experience extremely large delays. The distribution is positively skewed. Significant delays are less frequent but can affect connecting flights.

Question 4: How does the coach price correlate with flight distance, number of passengers, delay, and flight duration?

Solution

Python Code

```
corr = sample[['coach_price', 'miles', 'passengers', 'delay', 'hours']].corr()  
print(corr)
```

R Code

```
cor(sample[,c('coach_price', 'miles', 'passengers', 'delay', 'hours')])
```

Result

Table 1: Correlation Matrix					
	coach_price	miles	passengers	delay	hours
coach_price	1.000	0.344	0.168	-0.012	0.336
miles	0.344	1.000	-0.004	-0.011	0.985
passengers	0.168	-0.004	1.000	-0.008	-0.003
delay	-0.012	-0.011	-0.008	1.000	-0.012
hours	0.336	0.985	-0.003	-0.012	1.000

Interpretation

Coach ticket prices increase with flight distance and flight duration. Delay has almost no relationship with ticket prices.

Question 5: What is the relationship between coach and first-class prices? Do flights with higher coach prices always have higher first-class prices as well?

Solution

Python Code

```
sample[['coach_price', 'firstclass_price']].corr()

plt.scatter(sample['coach_price'], sample['firstclass_price'])
plt.xlabel('Coach Price')
plt.ylabel('First Class Price')
plt.title('Coach vs First Class Price')
plt.show()
```

R Code

```
cor(sample$coach_price, sample$firstclass_price)

plot(sample$coach_price, sample$firstclass_price,
      xlab='Coach Price', ylab='First Class Price')
```

Result

Correlation = 0.7493722

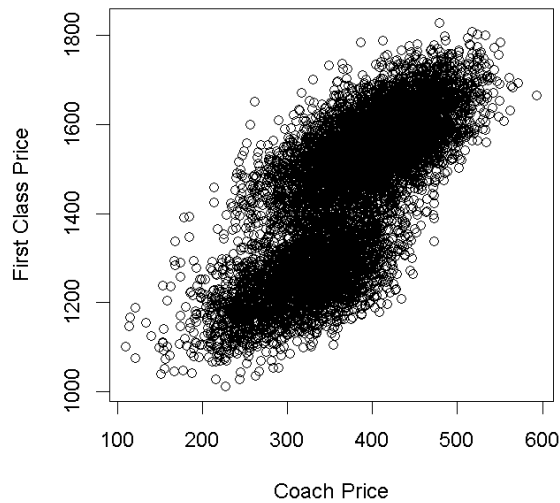


Figure 2: Plot

Interpretation

There is a strong positive relationship between coach and first-class ticket prices. Flights with expensive coach tickets usually also have expensive first-class tickets.

Question 6: What is the relationship between coach prices and in-flight features?

Solution

Python Code

```
# Average coach price by WiFi
wifi_result = sample.groupby('inflight_wifi')['coach_price'].mean()
print("Average Coach Price by WiFi")
print(wifi_result)

# Average coach price by Meal
meal_result = sample.groupby('inflight_meal')['coach_price'].mean()
print("\nAverage Coach Price by Meal")
print(meal_result)

# Average coach price by Entertainment
ent_result = sample.groupby('inflight_entertainment')['coach_price'].mean()
print("\nAverage Coach Price by Entertainment")
print(ent_result)
```

R Code

```
rm(mean)
aggregate(coach_price ~ inflight_wifi, sample, mean)
aggregate(coach_price ~ inflight_meal, data = sample, FUN = base::mean)
aggregate(coach_price ~ inflight_entertainment, data = sample, FUN = base::mean)
```

Result

Table 2: Average Coach Price Based on Inflight Services

Service	No	Yes
Inflight WIFI	314.9042	383.2341
Inflight meal	370.0618	390.4319
Inflight entertainment	319.2038	390.4368

Interpretation

Flights that provide in-flight Wi-Fi, meals, and entertainment systems generally have higher coach ticket prices. Among these features, in-flight entertainment is associated with the highest increase in coach ticket price. This suggests that passengers are willing to pay more for additional comfort and services during flights.

Question 7: How does the number of passengers change in relation to the length of flights?

Solution

Python Code

```
plt.scatter(sample['hours'], sample['passengers'])
plt.xlabel('Hours')
plt.ylabel('Passengers')
plt.title('Passengers vs Flight Duration')
plt.show()
```

R Code

```
plot(sample$hours, sample$passengers,
      xlab='Hours', ylab='Passengers')
```

Result

Figure 3: Passengers vs Flight Duration

Interpretation

Passenger counts do not change dramatically with flight duration. However, medium and long flights tend to carry slightly more passengers.

Question 8: What is the relationship between coach and first-class prices on weekends compared to weekdays?

Solution

Python Code

```
sample.groupby('weekend')[['coach_price', 'firstclass_price']].mean()
```

R Code

```
aggregate(cbind(coach_price, firstclass_price) ~ weekend, sample, mean)
```

Result

Table 3: Average Ticket Prices on Weekdays and Weekends

Weekend	Coach Price	Firstclass Price
No	322.7567	1259.806
Yes	405.0000	1558.844

Interpretation

Weekend flights generally have slightly higher ticket prices because travel demand is often higher during weekends.

Question 9: How do coach prices differ for redeyes and non-redeyes on each day of the week?

Solution

Python Code

```
sample.groupby(['day_of_week', 'redeye'])['coach_price'].mean()
```

R Code

```
aggregate(coach_price ~ day_of_week + redeye, sample, mean)
```

Result

Table 4: Average Coach Price by Day of Week and Redeye Flight

Day of Week	Redeye	Coach Price
Friday	No	403.66
Monday	No	326.25
Saturday	No	415.13
Sunday	No	405.02
Thursday	No	324.82
Tuesday	No	324.63
Wednesday	No	327.86
Friday	Yes	325.65
Monday	Yes	259.17
Saturday	Yes	350.44
Sunday	Yes	337.58
Thursday	Yes	249.42
Tuesday	Yes	258.92
Wednesday	Yes	251.96

Interpretation

Redeye flights are generally cheaper than non-redeye flights because many passengers prefer daytime travel.

Question 10: Perform the analysis for appropriate variables: Summary statistics, Visualization using histograms, and Hypothesis Testing

Solution

1.Independent t-test to Compare Group Means

Python Code

```
from scipy import stats
weekend_yes = sample[sample['weekend'] == 'Yes']['coach_price']
weekend_no = sample[sample['weekend'] == 'No']['coach_price']

print(stats.ttest_ind(weekend_yes, weekend_no))

import matplotlib.pyplot as plt
plt.hist(sample['coach_price'], bins=30)
plt.title("Histogram of Coach Ticket Prices")
plt.xlabel("Coach Price")
plt.ylabel("Frequency")
plt.show()
```

R Code

```
t.test(coach_price ~ weekend, data = sample)

hist(sample$coach_price,
      main="Histogram of Coach Ticket Prices",
      xlab="Coach Price", col="lightblue")
```

Result

Metric	Value
t-statistic	-71.644
Degrees of Freedom (df)	7264.1
p-value	$< 2.2 \times 10^{-16}$
95% Confidence Interval	[-84.60, -80.09]

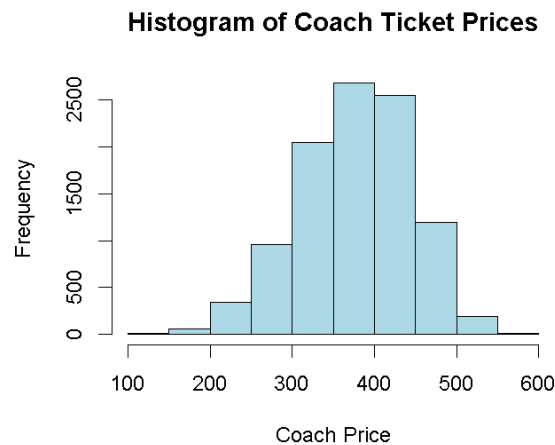


Figure 4: Histogram

Interpretation

The extremely small p-value ($< 2.2 \times 10^{-16}$) indicates a statistically significant difference between coach prices on weekends compared to weekdays. We reject the null hypothesis and conclude that weekend flight pricing behavior shifts substantially higher.

2.Price Differences Between Weekend and Weekday Flights

Solution

Python Code

```
sample.groupby('weekend')['coach_price'].mean()
plt.hist(sample[sample['weekend']=='Yes']
         ['coach_price'],
         bins=30)
```

```

plt.title("Weekend Flight Prices")
plt.xlabel("Coach Price")
plt.ylabel("Frequency")
plt.show()

plt.hist(sample[sample['weekend']=='No']
         ['coach_price'],
         bins=30)

plt.title("Weekday Flight Prices")
plt.xlabel("Coach Price")
plt.ylabel("Frequency")
plt.show()

```

R Code

```

aggregate(coach_price ~ weekend,
          sample, mean)
hist(sample$coach_price[sample$weekend=="Yes"],
     main="Weekend Flight Prices",
     xlab="Coach Price",
     col="lightgreen")

hist(sample$coach_price[sample$weekend=="No"],
     main="Weekday Flight Prices",
     xlab="Coach Price",
     col="lightblue")

```

Result

Table 5: Average Coach Ticket Prices for Weekend and Weekday Flights

Weekend	Coach Price
No	322.7567
Yes	405.1003

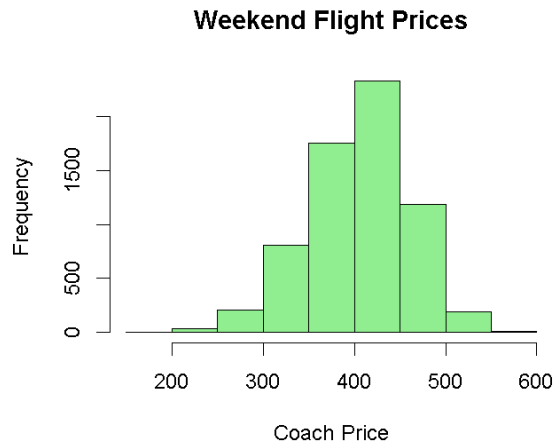


Figure 5: Weekend Flight Prices

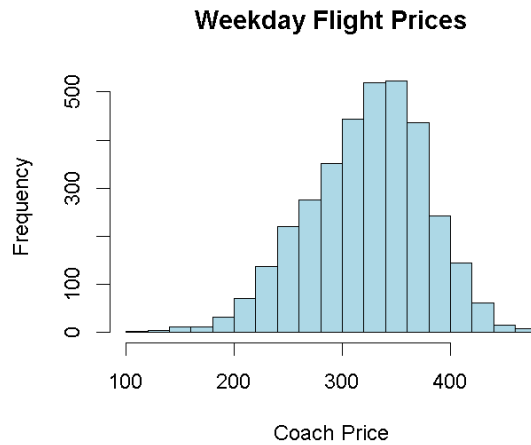


Figure 6: Weekday Flight Prices

Interpretation

Passenger demand usually increases during weekends because more people travel for vacations and personal purposes. Increased demand causes airlines to charge higher ticket prices.

3. Correlation Analysis Among Numeric Variables

Solution

Python Code

```
sample[['coach_price', 'miles', 'passengers', 'delay', 'hours']].corr()
plt.hist(sample['miles'],
         bins=30)
```

```
plt.title("Histogram of Miles")
plt.xlabel("Miles")
plt.ylabel("Frequency")
plt.show()

plt.hist(sample['hours'],
         bins=30)

plt.title("Histogram of Flight Hours")
plt.xlabel("Hours")
plt.ylabel("Frequency")
plt.show()
```

R Code

```
cor(sample[,c('coach_price', 'miles', 'passengers', 'delay', 'hours')])
hist(sample$miles,
     main="Histogram of Miles",
     xlab="Miles",
     col="orange")

hist(sample$hours,
     main="Histogram of Flight Hours",
     xlab="Hours",
     col="pink")
```

Result

Table 6: Correlation Matrix					
	coach_price	miles	passengers	delay	hours
coach_price	1.000	0.344	0.168	-0.012	0.336
miles	0.344	1.000	-0.004	-0.011	0.985
passengers	0.168	-0.004	1.000	-0.008	-0.003
delay	-0.012	-0.011	-0.008	1.000	-0.012
hours	0.336	0.985	-0.003	-0.012	1.000

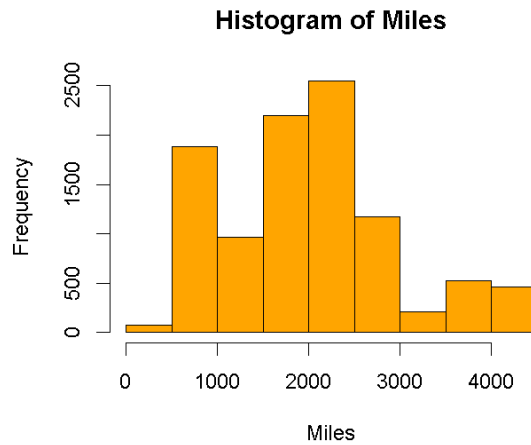


Figure 7: Histogram of Miles

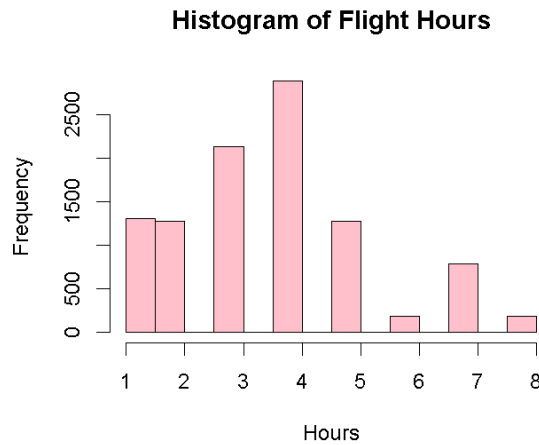


Figure 8: Histogram of Flight Hours

Interpretation

Long-distance and long-duration flights generally have higher ticket prices. Flight delays have very little influence on airline ticket pricing.

4. Regression Analysis:

Solution

Python Code

```
import statsmodels.formula.api as smf
model = smf.ols('coach_price ~ miles + passengers + delay + hours',data=sample).fit()
print(model.summary())
plt.hist(sample['delay'],
         bins=30)
```

```
plt.title("Histogram of Flight Delays")
plt.xlabel("Delay")
plt.ylabel("Frequency")
plt.show()
```

R Code

```
model <- lm(coach_price ~ miles + passengers +delay +hours,data = sample)
summary(model)
hist(sample$delay,
      main="Histogram of Flight Delays",
      xlab="Delay",
      col="yellow")
```

Result

Table 7: Regression Analysis Results

Predictor	Estimate	Std. Error	t value	Pr(> t)	Significance
(Intercept)	140.5630	10.2248	13.747	$< 2 \times 10^{-16}$	***
miles	0.0302	0.0039	7.743	1.06×10^{-14}	***
passengers	0.8959	0.0487	18.401	$< 2 \times 10^{-16}$	***
delay	-0.0137	0.0183	-0.744	0.457	
hours	-2.9055	2.1192	-1.371	0.170	

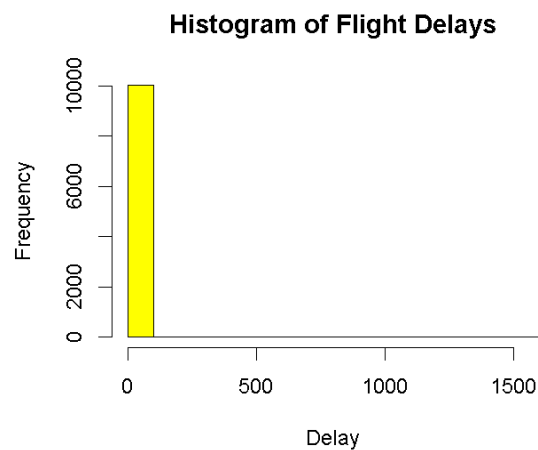


Figure 9: Histogram of Flight Delays

Interpretation

The regression model shows that miles travelled and number of passengers are the primary factors affecting coach ticket prices, and both variables are highly statistically significant

($p < 0.001$). The model explains approximately 14.7% of the variation in coach ticket prices. The remaining variation may be influenced by other factors such as weekend travel, redeye flights, in-flight services, and seasonal demand.

For every additional 100 miles travelled, the coach ticket price increases by approximately \$3.02. Similarly, each additional passenger is associated with an increase of approximately \$0.90 in the coach ticket price.

Delay and flight duration are not statistically significant because their p-values are greater than 0.05, indicating that these variables do not strongly affect coach ticket prices in this regression model.

5. Linear Regression to Predict Ticket Prices

Solution

Python

```
linear_model = smf.ols('coach_price ~ miles + hours',data=sample).fit()
print(linear_model.summary())
plt.hist(sample['coach_price'],
         bins=30)

plt.title("Distribution of Coach Prices")
plt.xlabel("Coach Price")
plt.ylabel("Frequency")
plt.show()
```

R Code

```
linear_model <- lm(coach_price ~ miles + hours,data = sample)
summary(linear_model)
hist(sample$coach_price,
     main="Distribution of Coach Prices",
     xlab="Coach Price",
     col="lightblue")
```

Result

Table 8: Linear Regression Results for Predicting Coach Ticket Prices

Variable	Estimate	Std. Error	t value	Pr(> t)	Significance
(Intercept)	326.4614	1.5004	217.580	$< 2 \times 10^{-16}$	***
Miles	0.0299	0.0039	7.550	4.74×10^{-14}	***
Hours	-2.7824	2.1545	-1.291	0.197	

Table 9: Linear Regression Model Summary

Statistic	Value
Observations	10,033
Residual Standard Error	63.56
Multiple R-squared	0.1182
Adjusted R-squared	0.1180
F-statistic	672.1
p-value	$< 2.2 \times 10^{-16}$

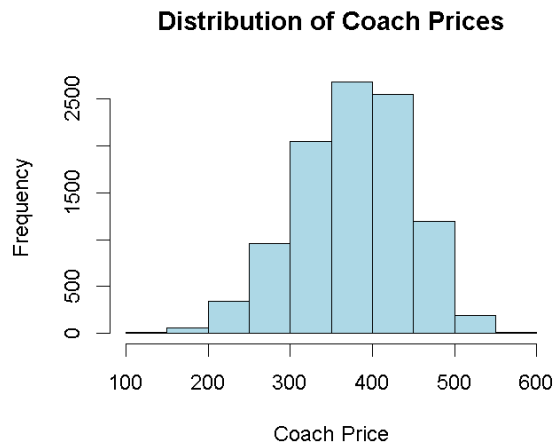


Figure 10: Distribution of Coach Price

Interpretation

Miles travelled is the only statistically significant predictor in this linear regression model ($p < 0.001$). This indicates that flight distance is the primary factor affecting coach ticket prices.

The model explains approximately 11.8% of the variation in ticket prices, based on the Multiple R^2 value. Although the model is statistically significant, a large portion of ticket price variation is still influenced by other factors not included in this model.

The very high correlation between Miles and Hours (0.985) suggests strong multicollinearity between these variables. Since both variables provide nearly the same information, adding flight duration (Hours) to the model does not contribute much additional predictive power once Miles is already included.

For every additional 100 miles travelled, the coach ticket price increases by approximately \$2.99.

6. Logistic Regression for Binary Outcomes (Redeye Flights)

Solution

Python Code

```
import numpy as np
```

```

sample['redeye_binary'] = np.where(sample['redeye'] == 'Yes', 1, 0)
log_model = smf.logit('redeye_binary ~ hours + miles + coach_price',data=sample).fit()
print(log_model.summary())
plt.hist(sample['hours'],
         bins=30)

plt.title("Histogram of Flight Duration")
plt.xlabel("Hours")
plt.ylabel("Frequency")
plt.show()

```

R Code

```

sample$redeye_binary <-ifelse(sample$redeye == "Yes", 1, 0)
log_model <- glm(redeye_binary ~ hours + miles + coach_price,data = sample,family = b
summary(log_model)
hist(sample$hours,
     main="Histogram of Flight Duration",
     xlab="Hours",
     col="lightgreen")

```

Result

Table 10: Logistic Regression Results for Redeye Flights

Predictor	Estimate	Std. Error	z value	Pr(> z)	Significance
(Intercept)	1.7103	0.2301	7.432	1.07×10^{-13}	***
Hours	-0.1104	0.1619	-0.682	0.4952	
Miles	0.0006	0.0003	2.009	0.0445	*
Coach Price	-0.0157	0.0007	-21.369	$< 2 \times 10^{-16}$	***

Table 11: Logistic Regression Model Summary

Metric	Value
Null Deviance	3967.8 (on 10,032 DF)
Residual Deviance	3464.4 (on 10,029 DF)
AIC	3472.4
Fisher Iterations	6

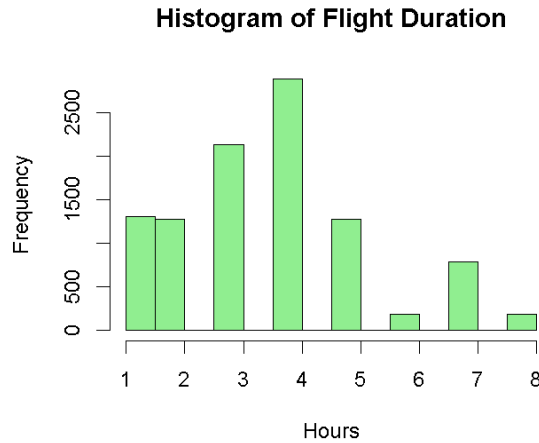


Figure 11: Enter Caption

Interpretation

- Coach Price (Primary Driver):** There is a highly significant negative relationship between coach ticket price and redeye flight status ($p < 2 \times 10^{-16}$). For every one-dollar increase in coach ticket price, the likelihood (log-odds) of the flight being a redeye decreases by approximately 0.0157. This confirms that redeye flights are generally priced lower than daytime flights.
- Distance (Miles):** The variable *miles* is statistically significant at the 0.05 significance level ($p = 0.0445$). This suggests that longer-distance flights are slightly more likely to operate as redeye flights.
- Flight Duration (Hours):** Flight duration is not statistically significant ($p = 0.4952$). After controlling for distance and coach ticket price, flight duration does not independently predict whether a flight is a redeye flight.
- Model Fit (AIC and Deviance):** The model has an AIC value of 3472.4. The Residual Deviance (3464.4) is lower than the Null Deviance (3967.8), indicating that including the predictor variables improves the model's ability to predict redeye flights compared to a model without predictors.